

Romi Lesson 19 Update to RomiBot2026

At this point, you probably have about 90% of what you need to know to program the competition robot. The lessons up to this point have used the RomiReference code as a starting point. This code was written originally several years ago and as you have seen, it includes many cases of deprecated functions and other issues. From this point forward we will start our lessons with a more up to date program. This code is much closer to the competition configuration than we have been using.

You need to clone RomiBot2026 from GitHub onto your computer.

(<https://github.com/team294/RomiBot2026>) Once you do that, open a new branch with your name on it which will be your workspace.

There are several new concepts in this code that you need to get familiar with. This code is compatible with the 2026 FRC competition code. (There will be some changes for the 2027 season since the RoboRio is being replaced, but these are not available as of this writing.)

A subsystem LED.java was added to control Leds separate from OnBoardIO. This enables subsystem methods like LED.get() etc. which were not available in the RomiReference code OnBoardIO which we had been using.

Data Logging is an important function for improving and troubleshooting the robot. By reviewing the robot log after a match or practice run, we gain incite into how the systems performed and how they can be improved.

Several other utilities are added to add some special math functions and data logging functions that are not included in wpilib.

The code is more robust and readable with better adherence to putting all fixed constants in the constants class.

Read through this code to be familiar with all the classes and structure.

To understand this code, we will do two exercises:

Exercise 1: In your new branch with labeled with your name,

Task: Update your RomiBot2024 branch from remote Main

Task: Add a green LED to the LED subsystem

Task: Add Elastic buttons to turn green LED on and off. Test and debug your code. Save the Elastic layout to the RomiBot2024 project folder and commit.

Exercise 2:

Task: Add a get method to the LED subsystem.

Task: Add a command to toggle the green LED.

Task: Use this get method to determine if the led needs to be turned on or off.

Task: Add a widget to Elastic to toggle the green LED.

Task: Test. Save. Commit

Exercise 3: Optional for extra credit

Task: Modify the ToggleLed command to pass the color parameter, so any of the three leds will toggle. Reference the LEDSet command.

When finished, commit and push to GitHub.